LVAD Emergency

AC-14 | Adult Cardiac

Quick flow

01 Unresponsive LVAD Patient [EMT / AEMT / Paramedic]

Support ventilation as needed • do not diagnose arrest from pulse absence alone

02 Assess Perfusion [EMT / AEMT / Paramedic]

Skin color/temperature • capillary refill • Doppler MAP • waveform $EtCO_2$

03 Adequate Perfusion? [EMT / AEMT / Paramedic]

Any present: normal skin • normal capillary refill • MAP >50 mmHg • $EtCO_2$ >20 mmHg

04 Yes - No Compressions [EMT / AEMT / Paramedic]

Treat complaint and non-LVAD causes • monitor perfusion • contact LVAD center • transport

05 No - Start CPR Now [EMT / AEMT / Paramedic]

Begin compressions immediately • follow AC-03/AC-09 • do not delay for device troubleshooting

06 Second Rescuer: Check LVAD [EMT / AEMT / Paramedic]

Listen for hum • read alarms • inspect driveline/controller • restore connected power

07 Transport + Coordinator [EMT / AEMT / Paramedic]

LVAD-capable/implanting center when appropriate • bring caregiver and all backup equipment

Overview

- Emergency assessment and treatment pathway for an adult with a durable continuous-flow left ventricular assist device outside the immediate implant hospitalization.
- Continuous-flow LVAD patients may not have a palpable pulse. Determine the need for resuscitation from responsiveness and perfusion rather than pulse absence alone.
- For an unresponsive patient with impaired perfusion, begin chest compressions immediately while a second rescuer evaluates the LVAD and corrects device-related causes.

Provider-Level Actions

All provider levels assess perfusion and begin lifesaving BLS care immediately; device evaluation occurs concurrently when personnel permit.

EMT

- Assess responsiveness, breathing, skin color and temperature, capillary refill, and other available signs of perfusion; pulse absence alone does not indicate arrest.
- Support ventilation, apply AED/monitor pads, and begin chest compressions immediately when the patient is unresponsive with impaired perfusion.
- Listen for the LVAD hum, read controller alarms, inspect visible connections, restore an obvious disconnected power or driveline connection, and gather all backup equipment.

AEMT

- Perform all EMT actions plus IV/IO access, cautious isotonic-fluid aliquots for impaired perfusion with suspected hypovolemia, and frequent reassessment.
- Use smaller aliquots and reassess closely when right-ventricular failure, pulmonary edema, or volume overload is possible.
- Do not delay CPR, defibrillation, device troubleshooting by a second rescuer, coordinator contact, or transport for vascular access.

Paramedic

- Perform all AEMT actions plus rhythm interpretation, manual defibrillation or synchronized cardioversion when indicated, advanced airway management, and waveform capnography.
- Use Doppler with a manual cuff when available, treat dysrhythmia and reversible causes under the linked protocols, and trend perfusion after each intervention.
- Coordinate the LVAD-center report, Medical Control consultation, destination decision, and any termination evaluation.

Indications

- Adult with a durable continuous-flow LVAD who is ill, injured, unresponsive, poorly perfused, or experiencing a device alarm or suspected device malfunction.
- This pathway applies beyond the immediate LVAD implant hospitalization.

Assessment | Perfusion assessment

- Assess responsiveness and breathing; support ventilation immediately when inadequate.
- Do not use absence of a palpable pulse alone to diagnose cardiac arrest because continuous-flow LVAD patients commonly have no palpable pulse.
- Perfusion is adequate when any of the following is present: normal skin color and temperature, normal capillary refill, MAP greater than 50 mmHg, or $EtCO_2$ greater than 20 mmHg.
- An unresponsive LVAD patient without any adequate-perfusion criterion is likely in cardiac arrest.
- Automated blood-pressure and pulse-oximetry measurements may be unreliable. Use Doppler with a manual cuff when available and trend waveform capnography when appropriate.

Assessment | LVAD and equipment assessment

- Identify the device and implanting center from the emergency card, controller, equipment labels, caregiver, or medical alert information.
- Listen for a continuous mechanical hum over the left chest and left upper abdomen.
- Check the controller display and alarms, visible driveline connection and damage, battery charge, cable connections, and external power source.
- Reconnect an obvious disconnected driveline, battery, or power cable when it can be done without force or device disassembly.
- Ask the patient or trained caregiver what the alarm means, the patient's usual MAP and pump readings, and whether the current equipment differs from baseline.

Assessment | Concurrent cause assessment

- Evaluate for dysrhythmia, hemorrhage or hypovolemia, hypoxia, hypoglycemia, overdose, right-ventricular failure, sepsis, stroke, trauma, and another time-sensitive complaint.
- Identify anticoagulant and antiplatelet medications, last dose when known, bleeding, melena or hematemesis, recent falls, and focal neurologic symptoms.
- A ventricular tachyarrhythmia may be tolerated while the patient remains conscious and perfused; treatment is determined by symptoms and perfusion rather than the monitor alone.

Treatment steps

- Support ventilation as needed and determine whether adequate perfusion is present; do not delay immediate lifesaving care while locating device information.
- When any adequate-perfusion criterion is present, do not perform chest compressions. Monitor closely and treat the complaint and non-LVAD causes under the applicable linked protocol.
- When the patient is unresponsive with impaired perfusion, begin chest compressions immediately and follow AC-03 and the applicable rhythm pathway.

- When a second rescuer is available, assess the LVAD concurrently: listen for hum, read alarms, inspect the driveline and controller, reconnect an obvious disconnection, and ensure adequate power.
- Replace depleted batteries one at a time so that one power source remains connected at all times.
- A controller exchange may be performed only by a trained patient or caregiver or under direct instruction from the LVAD coordinator.
- Perform defibrillation or synchronized cardioversion when indicated without disconnecting the LVAD; position pads away from the implanted pump, controller, and driveline.
- Control external hemorrhage aggressively. Give cautious isotonic-fluid aliquots only when impaired perfusion and suspected hypovolemia are present, reassessing for pulmonary edema, right-ventricular failure, or volume overload.
- Contact the LVAD coordinator or implanting center early, but do not delay CPR, treatment, or transport. Consult Medical Control for destination or complex treatment decisions.
- Transport to an LVAD-capable or implanting center when clinically and geographically appropriate. Bring the caregiver, backup controller, charged batteries, charger or power module, emergency card, device records, and medications.

Medications

Anticoagulant / antiplatelet consideration

No routine prehospital reversal under this protocol

- Expect chronic anticoagulant or antiplatelet therapy and control external hemorrhage aggressively.
- Use a separate approved reversal policy or direct Medical Control order when reversal is considered.

Isotonic crystalloid

Cautious aliquots for impaired perfusion with suspected hypovolemia

- Reassess MAP, lung sounds, work of breathing, skin perfusion, $EtCO_2$, and response after each aliquot.
- Use smaller aliquots when right-ventricular failure, pulmonary edema, or volume overload is suspected.

Warnings / precautions

- Do not use pulse absence alone to diagnose cardiac arrest in a continuous-flow LVAD patient.
- Do not delay chest compressions in an unresponsive patient with impaired perfusion while troubleshooting the device or contacting the coordinator.
- Never disconnect both LVAD power sources simultaneously.
- Do not cut, clamp, pull, or otherwise manipulate the driveline and do not force damaged connections.
- Do not exchange the controller unless the patient or caregiver is trained or the LVAD coordinator directly instructs the exchange.
- Do not withhold indicated defibrillation or synchronized cardioversion solely because an LVAD is present.
- Do not perform routine prehospital anticoagulant reversal without a separate approved policy or direct Medical Control order.
- Do not terminate resuscitation based solely on pulse absence, device alarm, rhythm, or $EtCO_2$.
- Destination selection must account for patient stability, transport time, LVAD-center capability, and coordinator or Medical Control direction.

Clinical pearls

- A functioning continuous-flow LVAD usually produces a constant mechanical hum over the left chest or left upper abdomen.
- Normal skin findings, normal capillary refill, MAP greater than 50 mmHg, or $EtCO_2$ greater than 20 mmHg each independently indicate adequate perfusion in the AHA durable-LVAD algorithm.
- Current AHA guidance prioritizes immediate CPR for an unresponsive LVAD patient with impaired perfusion; a second rescuer should troubleshoot the device in parallel.

- Automated blood pressure and pulse oximetry may fail in low-pulsatility flow; Doppler MAP and waveform $EtCO_2$ can provide more useful perfusion information.
- Bring the patient's trained caregiver whenever possible because the caregiver may know the alarm, normal readings, and emergency equipment better than responding clinicians.

Special populations | Perfusing ventricular tachyarrhythmia

- A conscious, adequately perfused LVAD patient may tolerate VT or VF; do not begin chest compressions solely from the ECG rhythm.
- Treat symptomatic or poorly perfused tachyarrhythmia under the applicable rhythm protocol and contact the LVAD center early.

Special populations | Trauma or bleeding

- Assume anticoagulant or antiplatelet use until verified, control external hemorrhage immediately, and maintain a low threshold for serious occult bleeding or intracranial injury.
- Use the applicable trauma pathway without interrupting essential LVAD power or damaging the driveline.

Special populations | Termination or advance directive

- Honor a valid DNR, MOST, or advance directive under agency policy and use separate obvious-death criteria when applicable.
- Otherwise, termination requires AC-12 criteria, Medical Control authorization, and LVAD-coordinator consultation when the coordinator is reachable.

Related pathways / contacts

AC-03 Adult Cardiac Arrest - Open immediately for an unresponsive LVAD patient with impaired perfusion.

AC-09 VF / Pulseless VT - Open for shockable VF/pVT arrest treatment when perfusion is absent.

AC-10 Post-Resuscitation Care - Open immediately after sustained ROSC or restoration of adequate perfusion.

AC-11 TEAM-Focused CPR - Open for concurrent CPR-team roles, timing, and quality management.

AC-12 Termination of Resuscitation - Open when termination is being considered; Medical Control and LVAD consultation are required as specified above.

UP-04 Altered Mental Status - Open for hypoglycemia or another non-LVAD altered-mental-status cause.

TE-07 Overdose / Toxic Ingestion - Open for suspected overdose or toxicologic cause.

TB-10 Traumatic Arrest - Use for LVAD patients in traumatic cardiac arrest.

Source and references

American Heart Association. 2025 Guidelines for CPR and ECC: Adult and Pediatric Special Circumstances of Resuscitation - Left Ventricular Assist Devices. | American Heart Association. 2025 Adult and Pediatric Durable LVAD Algorithm. | Abbott. HeartMate 3 LVAD system and patient emergency-equipment guidance. | Claiborne EMS adult cardiac-arrest, rhythm, airway, altered-mental-status, toxicology, trauma, and termination protocols.

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